

Building Technology

Engineering Mathematics I

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$y = mx + c$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\tan \theta = \frac{\text{rise}}{\text{run}}$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$V = \frac{1}{3}Ah$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$C = \pi d$$

Section B

5. Calculate roof pitch angle. **(10 marks)**
6. Find volume of excavation. **(10 marks)**
7. Interpret a straight-line graph from site data. **(10 marks)**
8. Create one realistic building technology problem based on this topic and solve it fully. **(20 marks)**